# The Middle East Water Crisis

## GERD, Desalination, and the World's Most Water-Stressed Region

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## Executive Summary

The Middle East and North Africa region faces structural water scarcity driven by aridity, growing populations, and accelerating climate change. The Arabian Peninsula's Gulf states are structurally dependent on desalination and fossil aquifer extraction to sustain modern urban populations. Water insecurity intersects directly with geopolitical conflict: Iran's threats in early 2026 explicitly included Gulf water infrastructure alongside energy facilities as potential targets of retaliation, underscoring how deeply water systems are entangled with regional security. The following sections assess the state of the crisis using available evidence.

## Part I — Climate Change and Water Stress in the Middle East

The IPCC Sixth Assessment Report documents that climate change is projected to exacerbate risk and add new vulnerabilities for water systems across regions including the Middle East. Rising atmospheric $CO_2$ concentration is projected to decrease water efficiency of growing crops in parts of the Mediterranean region. Research cited in the AR6 corpus — including Barlow et al. (2015) on drought in the Middle East and Central-Southwest Asia and a subsequent 2016 review of drought in the Middle East and Southwest Asia published in the Journal of Climate — establishes that drought conditions in the region are a documented and worsening phenomenon. Heat-stress mortality risk due to global warming in the Middle East and North Africa (MENA) has been identified as an escalating concern in peer-reviewed literature.

Adaptation to water-related risks and impacts makes up the majority of all documented adaptation responses globally, according to the IPCC AR6. A large number of these responses are in the agriculture sector and include on-farm water management, water storage, soil moisture conservation, and irrigation. The ability to adapt using irrigation will be increasingly limited by water availability, especially at higher warming levels.

Land-use change and water extraction for irrigation have influenced local and regional responses in the water cycle. Both importing and exporting countries are exposed to transboundary risk transmission through climate change impacts on distant water resources. Climate change is projected to exacerbate this risk and add new vulnerabilities for transboundary risk transmission.

## Part II — Gulf Water Infrastructure as a Geopolitical Target

In March 2026, the intersection of water security and armed conflict became direct and explicit. Iran threatened to retaliate against Gulf energy and water infrastructure following a Trump ultimatum. Trump threatened to hit Iran's electricity grid if the country did not fully open the Strait of Hormuz. CNA[2026-03-23]

Iran's strategic logic in attacking Gulf infrastructure was assessed by analysts as a deliberate effort to increase the costs of conflict even to parties not directly involved. Iran attacked energy facilities across the Gulf, including QatarEnergy's liquefied natural gas production facilities in Ras Laffan Industrial City, Qatar. Tehran's strategy of targeting Gulf infrastructure — including the explicit threat to water systems — signals the vulnerability of desalination-dependent states in any regional military escalation. CNA[2026-03-21]

The inclusion of water infrastructure in Iran's stated threat envelope is analytically significant. Gulf states derive their municipal water supply from desalination plants concentrated along coastlines — the same geography as LNG terminals and energy facilities. Any sustained attack on these installations would threaten the water supply of populations with no alternative freshwater source.

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## Part III — Regional Geopolitics: The Jordan-Israel Dimension

Israel and Jordan maintain a bilateral relationship that encompasses both security and water dimensions. In January 2014, Israeli Prime Minister Benjamin Netanyahu travelled to Amman for talks with King Abdullah II on the US-brokered Middle East peace process. The visit was unannounced and followed extensive US diplomatic engagement in the region. CNA[2014-01-20]

Qualitative evidence of ongoing political engagement between Israel and Jordan over shared resources exists in the diplomatic record. However, specific figures for water-sharing volumes from the evidence bundle are absent. No quantitative claims about Jordan River allocations or specific cubic-metre transfer commitments can be made on the basis of available evidence.

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## Part IV — Gulf Energy-Water Linkage and ASEAN Parallels

ASEAN discussions in May 2026 highlighted the strategic importance of Gulf nations as partners for regional resilience frameworks. Analysts recommended that ASEAN work with Gulf nations to establish decentralised strategic stockpiles, noting that physical capacities in Southeast Asia were limited. The proposal would involve the private sector and enable member states to access shared reserves. Malaysia Prime Minister Anwar Ibrahim called for ASEAN to "fully leverage" cooperation with Gulf nations and China under existing mechanisms to build more reliable and resilient energy arrangements. CNA[2026-05-11]

While this discussion centred on fuel and energy, the strategic logic applies to water with equal force: Gulf desalination infrastructure creates a concentrated point of failure in regional supply chains. The decentralised stockpile model proposed for energy is conceptually analogous to the water resilience challenge facing the Gulf.

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## Part V — Evidence Gaps

The following sections present in the original report but have no supporting evidence in the current evidence bundle and are therefore omitted from this rewrite:

- Saudi desalination capacity figures (volumes, plant counts, cost per cubic metre): no supporting chunks.
- UAE, Qatar, Kuwait desalination specifics: no supporting chunks.

- Grand Ethiopian Renaissance Dam (GERD) — reservoir capacity, hydropower generation, filling timeline: no supporting chunks. The query "Grand Ethiopian Renaissance Dam Nile Egypt" returned only H.G. Wells history text, Civilization VI game data, and unrelated Wikipedia content.
- Israel's water technology sector — Netafim, drip irrigation statistics, desalination plant capacities, wastewater recycling rates: no supporting chunks.
- Saudi fossil aquifer depletion — Wajid Aquifer age, extraction volumes, wheat self-sufficiency history, alfalfa ban: no supporting chunks.
- Libya's Great Man-Made River: no supporting chunks.
- Jordan River per capita availability figures: no supporting chunks.
- Egypt's Nile treaty allocation: no supporting chunks.
- Yemen aquifer depletion rates: no supporting chunks.

These claims were present in the prior version of this report but cannot be reproduced here because they do not appear in the evidence bundle. They should be sourced from BL3 MySQL or a dedicated MENA water RAG ingest before reinsertion.

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## Conclusion

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The Middle East water crisis is a documented structural emergency rooted in arid geography, population growth, and accelerating climate change. The IPCC AR6 confirms that drought risk in the Middle East is a well-established and worsening phenomenon. The March 2026 Iran-Gulf confrontation demonstrated that Gulf desalination infrastructure — the physical foundation of water supply for tens of millions of people on the Arabian Peninsula — is an explicit target in regional military escalation scenarios, not merely a civilian utility. CNA[2026-03-23]

The convergence of climate stress, fossil resource depletion, and geopolitical conflict over water systems constitutes a compound threat with no short-term resolution. Policy responses require energy-water nexus planning that accounts for the vulnerability of desalination infrastructure to both military attack and climate-driven energy disruption.

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